Information Geometry of Classical Carriers and Non-Classical Payloads in Biological Information Processing

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Abstract

Decoherence maps a d-level quantum system from a (d^2-1)-dimensional Bures state manifold onto a $(d-1)$-dimensional Fisher–Rao simplex, annihilating the d^2-d coherence directions. This dimensional collapse yields a continuous non-classicality index χ from Bures principal angles, allowing one to quantify how much coherence-sector structure remains geometrically accessible. We use that geometry to distinguish *classical carriers* from *non-classical payloads*. Photosynthetic exciton transport provides the proof of principle: validated FMO and PE545 models operate near the quantum-classical boundary, showing that biologically functional dynamics can inhabit a near-classical but non-classical regime. Neural oscillations, by contrast, are expected to be deep classical at carrier scale. A minimal KcsA-like ion-channel calculation then gives a concrete neural payload anchor: at an illustrative biological operating point $\gamma_{\text{bio}}/J_{\max} = 6.67$, the model remains boundary-near with $\theta_{\min} \approx 88.55^\circ$ and $\chi \approx 2.76 \times 10^{-4}$. For molecular modules i with local non-classical load $L_i = \chi_i(d_i^2 - d_i)$, the redundancy-adjusted payload $\mathcal{L}_{\text{eff}}(E) = \varrho_E^{-1} \sum_{i \in E} L_i$ is extensive whenever a nonzero fraction of modules sit near the boundary and redundancy is bounded. A minimal soft phase-coupling carrier model then shows how lower-frequency carriers can coordinate larger payload sets while remaining more classical themselves: coordinated dimensionality grows as the payload coordination radius expands and saturates at an amplitude ceiling. The resulting picture is a multi-scale architecture in which deep-classical macroscopic waves coordinate an extensive near-boundary molecular substrate. The present paper does not claim that this non-classical payload is already known to be functionally decisive for cognition; it establishes that the payload is nonzero, structured, extensive in coordinated site count, and empirically testable.

Keywords: decoherence; information geometry; quantum biology; neural oscillations; ion channels; dimensionality

1 Introduction

High-dimensional neural activity is often discussed as if it were primarily a property of fast dynamics. In the working-memory literature, gamma and beta bursts are closely associated with access, gating, and mixed selectivity [1, 2, 3, 4]. One frequency-centric reading of that literature supports a simple heuristic: high frequency goes with high dimensionality. This paper argues that the heuristic places dimensionality in the wrong object. Frequency is a temporal

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property of a carrier. Dimensionality is a property of a state space. Those are not the same thing.

This paper develops a different answer. The claim is not that low-frequency oscillations are themselves quantum, nor that high dimensionality automatically implies non-classicality. The claim is narrower and stronger: biological systems are plausibly organized as *carrier-payload architectures*. At macroscopic scale, oscillatory carriers can be deep classical. At molecular scale, the modules being coordinated can remain near the quantum-classical boundary. Dimensionality then scales with the coordinated payload, not with carrier frequency itself. The disagreement with a frequency-centric picture is therefore not about whether fast bursts matter. It is about *where the dimensionality resides*: in this paper, fast activity can index access or routing, while low-frequency carriers can dominate effective dimensionality by coordinating a larger payload.

The mathematical basis for that claim comes from decoherence geometry. Decoherence is the standard route by which quantum dynamics lose phase-sensitive distinguishability and approach classical pointer descriptions [5, 6, 7, 8]. A d-level quantum system occupies a (d^2-1)-dimensional state manifold under the Bures metric [9, 10, 11]. Complete dephasing collapses this manifold onto the $(d-1)$-dimensional classical probability simplex under the Fisher–Rao metric [12, 13]. The collapse is therefore not just loss of "quantumness" in an informal sense; it is a geometric destruction of distinguishable directions in state space.

That perspective is already useful in quantum biology. Photosynthetic complexes are known to operate in regimes where intermediate dephasing can improve transport efficiency [14, 15, 16, 17, 18]. What has been missing is a language for saying how classical or non-classical those biologically relevant operating points actually are. The Bures principal-angle framework supplies that language.

The broader motivation is cognitive. If one assumes that dimensionality is carried mainly by high-frequency activity, then low-frequency waves look low-dimensional by construction. If instead dimensionality is allowed to reside in the coordinated substrate, then the picture changes. A low-frequency wave can be *more* useful precisely because it is more classical: it is a reliable large-scale channel that can entrain many payload modules without introducing much quantum-classical mixing of its own. The core inversion is simple: high frequency may correlate with flexible access, but low frequency may carry more system-level dimensionality because it reaches farther and coordinates broader territories [19].

The paper develops that thesis in four steps. Section 2 states the core geometric facts of decoherence and defines the non-classicality index χ. Section 3 builds a carrier-payload formalism in which local module loads add into a redundancy-adjusted effective payload. Section 4 uses photosynthesis as proof of principle and then prioritizes ion channels as the leading neural payload candidate, with protein microdomains as a secondary extension. Section 5 introduces a minimal soft phase-coupling bridge from carrier frequency to coordinated payload size and states the resulting low-frequency/high-dimensional consequence for biological computation. The central outcome is a change of locus: frequency is recast as a coordination parameter, while dimensionality lives in the coordinated payload.

The burden of proof is therefore stratified. The geometric statements in Sections 2 and 3 are theorem-level claims. The photosynthetic anchors, ion-channel payload, protein microdomain payload, and reduced carrier proxies are model-based computations. The stronger neuroscience conclusion is intentionally narrower: the paper does not claim to have derived intelligence from quantum biology, only to have shown that a classical-carrier/non-classical-payload architecture is mathematically coherent, biologically anchored, and directly falsifiable.

2 Geometry of the Quantum-Classical Boundary

2.1 Dimensional collapse under dephasing

Let $\mathcal{M}_Q$ denote the manifold of $d \times d$ density matrices. For a full-rank system, $\dim \mathcal{M}_Q = d^2 - 1$: there are $d-1$ independent populations and $d^2 - d$ off-diagonal coherence directions. Fix a pointer basis $\{|k\rangle\}_{k=1}^d$ and let

$$\Lambda_{\mathrm{dec}}(\rho) = \sum_{k=1}^{d} |k\rangle\langle k|\, \rho\, |k\rangle\langle k| \tag{1}$$

be the complete dephasing channel.

Theorem 2.1 (Dimensional collapse)**.** *Under complete dephasing, the (d^2-1)-dimensional quantum manifold collapses onto the $(d-1)$-dimensional classical probability simplex:*

$$d^2 - 1 \longrightarrow d - 1. \tag{2}$$

Equivalently, the $d^2 - d$ off-diagonal coherence directions are annihilated, and the Bures metric on the image of Λ_{dec} restricts to the Fisher–Rao metric.

Proof. The image of Λ_{dec} is the set of diagonal density matrices, which is the classical simplex Δ_{d-1}. Any derivative in an off-diagonal direction vanishes after applying Λ_{dec}, so those directions drop out of the quantum Fisher information matrix. For a diagonal state $\rho = \mathrm{diag}(p_1, \ldots, p_d)$ and a tangent vector $\delta\rho = \mathrm{diag}(\delta p_1, \ldots, \delta p_d)$ with $\sum_k \delta p_k = 0$, the Bures/SLD metric reduces to $\sum_k (\delta p_k)^2/p_k$, i.e. the Fisher–Rao metric on the simplex [10, 11, 12]. Hence only $d-1$ distinguishable directions remain. □

Theorem 2.1 is the geometric core of the paper. It makes decoherence quantitative in the strongest possible sense: decoherence is a rank-reducing projection of state space, not merely a qualitative washing-out of phase information.

2.2 Principal angles and non-classicality

At a diagonal state, diagonal and off-diagonal tangent directions are orthogonal under the Bures metric. Away from the simplex they need not be orthogonal; the failure of orthogonality is exactly what it means for classical and quantum directions to remain mixed.

Definition 2.2 (Non-classicality index)**.** Let $\theta_1, \ldots, \theta_{d-1}$ be the Bures principal angles between the diagonal and off-diagonal tangent subspaces at a full-rank state ρ. Define

$$\chi(\rho) = \frac{1}{d-1} \sum_{a=1}^{d-1} \cos^2 \theta_a. \tag{3}$$

By construction, $\chi(\rho) \in [0, 1]$. Diagonal states have $\theta_a = \pi/2$ for all a, hence $\chi = 0$. Any state with geometric mixing between classical and coherence directions has $\chi > 0$. The index is deliberately modest: it is not an entanglement measure, not a coherence monotone, and not a claim that a system is "quantum" in any maximal sense. It is a scalar summary of how much of the coherence sector remains geometrically accessible. In the dynamical applications below, χ is evaluated on the full-rank regularized readout state produced by the corresponding Lindblad model at the stated operating point.

Remark 2.3. Large dimensionality alone does not imply non-classicality. A very large classical system can still satisfy $\chi = 0$. The claim developed below is therefore extensive rather than ontological: if a system contains many modules with small but nonzero local χ_i, the total payload can scale with system size.

3 Carrier-Payload Architecture

3.1 Local payload load

To move from single systems to biological assemblies, we distinguish the *carrier* from the *payload*. The carrier is the large-scale coordinating process. The payload is the set of local modules whose internal state spaces are being coordinated.

Definition 3.1 (Local non-classical load)**.** For module i with local dimension d_i and non-classicality χ_i, define

$$L_i = \chi_i(d_i^2 - d_i). \tag{4}$$

The term $d_i^2 - d_i$ is the size of the coherence sector that decoherence would destroy under full projection. Multiplying by χ_i gives a geometry-weighted effective load. When $\chi_i = 0$, the module contributes no non-classical payload. When $\chi_i > 0$, some fraction of that coherence sector remains geometrically relevant.

3.2 Redundancy-adjusted effective payload

Biological modules are not independent. If many modules are strongly redundant, naive linear addition overcounts the accessible payload. We therefore carry an explicit redundancy penalty.

Definition 3.2 (Effective payload)**.** Let E be a set of entrained modules and let $\varrho_E \geq 1$ be a redundancy factor. Define

$$\mathcal{L}_{\text{eff}}(E) = \frac{1}{\varrho_E} \sum_{i \in E} L_i \tag{5}$$

and the effective coordinated-dimension score

$$D_{\text{coord}}(E) = \sum_{i \in E} (d_i - 1) + \mathcal{L}_{\text{eff}}(E). \tag{6}$$

This separates classical capacity from non-classical uplift. The first term counts the surviving simplex dimensions. The second term counts the residual geometrically accessible coherence sector after redundancy compression. The sum should be read as an effective geometry score for accessible state-space richness, not as a literal manifold-dimension identity.

3.3 Extensive payload under nonzero module density

Proposition 3.3 (Extensive non-classical payload)**.** *Let E_N be an entrained set of N modules. Assume:*

(i) *there exist constants $L_\star > 0$ and $p > 0$ such that at least a fraction p of modules in E_N satisfy $L_i \geq L_\star$;*

(ii) *the redundancy factor is uniformly bounded, $\varrho_{E_N} \leq \varrho_\star$.*

Then

$$\mathcal{L}_{\text{eff}}(E_N) \geq \frac{pL_\star}{\varrho_\star} N, \tag{7}$$

and hence the non-classical payload is extensive in N.

Proof. At least pN modules contribute load at least $L_\star$, so $\sum_{i \in E_N} L_i \geq pNL_\star$. Dividing by the bounded redundancy factor gives the stated bound. □

Proposition 3.3 is the paper's main bridge from local molecular geometry to system-scale dimensionality. It does not say that large systems are automatically non-classical. It says that if a nonzero density of modules sits near the boundary, then the total accessible non-classical load grows with system size instead of saturating away.

3.4 Low-frequency advantage as a coordination statement

Corollary 3.4 (Low-frequency carrier advantage)**.** *Let $f_1 < f_2$ be two carrier frequencies with entrained sets $E(f_1)$ and $E(f_2)$. Suppose:*

(i) $|E(f_1)| \geq |E(f_2)|$*;*

(ii) *the local payload statistics and redundancy bounds remain comparable across the two sets.*

Then

$$D_{\text{coord}}(E(f_1)) \geq D_{\text{coord}}(E(f_2)). \tag{8}$$

Proof. Apply Proposition 3.3 with comparable module statistics and bounded redundancy. The coordinated dimensionality grows with the number of entrained payload modules, not with carrier frequency itself. □

Corollary 3.4 is the formal version of the paper's headline thesis. Low frequency is not valuable because it is intrinsically high-dimensional. It is valuable because it can coordinate a larger ensemble while staying close to a classical carrier limit.

4 Biological Anchors

4.1 Photosynthesis as proof of principle

Photosynthetic exciton transport is the cleanest biological proof of principle because the quantum-to-classical tradeoff is already functionally established [14, 15, 16, 17, 18]. In calculations based on the validated Hamiltonians of Adolphs–Renger and Novoderezhkin [20, 21], the transport optimum occurs in a near-classical regime rather than at maximal coherence. Averaging over the two standard initial-site choices, FMO gives $\gamma_{\text{opt}}/J_{\max} \approx 1.53$ with $\theta_{\min}^{\text{opt}} \approx 87.96°$, while PE545 gives $\gamma_{\text{opt}}/J_{\max} \approx 7.87$ with $\theta_{\min}^{\text{opt}} \approx 89.02°$. At the biological operating point, the scans give $\theta_{\min}^{\text{bio}} \approx 87.67°$ (FMO) and $89.02°$ (PE545). In local-load units $L = \chi(d^2 - d)$, those same biological points correspond to $L_{\text{FMO}}^{\text{bio}} \approx 2.28 \times 10^{-2}$ and $L_{\text{PE545}}^{\text{bio}} \approx 4.93 \times 10^{-3}$. This is exactly the regime the present framework is built to describe: biologically relevant function with $\chi > 0$ but principal angles already close to $\pi/2$.

The role of photosynthesis in this paper is strategic. It shows that the near-boundary regime is not empty rhetoric. Biology can already be shown to function there. That makes it plausible that other molecular biological systems may also carry small but nonzero local loads. Table 1 and Figure 1 summarize the current anchor set and, just as importantly, show the

System	Role	Type	$\gamma_{\rm bio}/J_{\rm max}$	$\theta_{\rm min}^{\rm bio}$	$\chi^{\rm bio}$	Use in draft
FMO	Proof point	Direct	1.14	87.67°	5.42×10^{-4}	Functional near-boundary photosynthesis anchor.
PE545	Proof point	Direct	7.88	89.02°	8.80×10^{-5}	Independent photosynthetic confirmation of the same band.
Ion channel	Neural payload	Direct	6.67	88.55°	2.76×10^{-4}	Molecular anchor for the primary neural payload route.
Prot. microd.	Payload ext.	Direct	7.69	88.21°	3.66×10^{-4}	Secondary payload anchor outside ion conduction.
Synth. carrier	Carrier	Proxy	50	89.78°	3.72×10^{-6}	Synthetic gamma-band carrier proxy, already near classical.
P-D carrier	Carrier	Proxy	50	89.37°	2.83×10^{-5}	Potjans–Diesmann cortical-column proxy, still more classical than the payload band.

Table 1: Anchor systems used in the present draft. All rows are computed directly in the present analysis. The photosynthetic models are reduced to the quantities this draft actually needs: biological-point and transport-optimum geometry from validated Hamiltonians. The neural rows report carrier proxies at $\gamma/J_{\rm max} = 50$ rather than the true biological limit $\gamma_{\rm bio}/J_{\rm max} \gg 1$; the latter only pushes the carriers further toward the classical limit.

split the paper needs: molecular payload anchors cluster in the same near-boundary band as the photosynthetic proof-of-principle calculations, while two neural carrier proxies already sit farther right and closer to the classical limit. The comparison is not only angular: the ion-channel local load $L_{\rm ion} \approx 3.31 \times 10^{-3}$ is of the same order as PE545 at the biological point, so the neural payload anchor is not being placed in a wholly alien regime.

The table is doing more work than a simple operating-point list. Its χ column makes the carrier-payload separation explicit: payload anchors cluster in a small-but-nonzero non-classical band around 10^{-4}, while the carrier proxies are already down at 3.72×10^{-6} and 2.83×10^{-5} by $\gamma/J_{\rm max} = 50$. The paper's central split is therefore not rhetorical; it is numerically visible in the anchor comparison itself.

4.2 Ion channels as the primary payload example

If the paper is meant to say something concrete about cognition, ion channels come first. They sit at the molecular interface between microscopic state transitions and population-level excitability; they exist in enormous copy number; and they offer a direct route from local non-classicality to large entrained site counts. Structurally, selectivity filters and gating assemblies are small-dimensional molecular machines with repeated motifs, constrained geometries, and biologically relevant couplings [22, 23]. In the present framework that makes them the cleanest neural payload candidate: small local d, plausible boundary-near energy scales, and a direct path to large N.

To make that claim concrete, we compute a minimal 4-site KcsA-like selectivity-filter anchor. Using site energies $(30, 45, 45, 30)\,\mathrm{cm}^{-1}$, nearest-neighbor coupling $J_{\rm max} = 30\,\mathrm{cm}^{-1}$, an illustrative protein-environment dephasing scale $\gamma_{\rm bio} = 200\,\mathrm{cm}^{-1}$, and trapping from S_1 to S_4, the biological operating point sits at $\gamma_{\rm bio}/J_{\rm max} = 6.67$ with

$$\theta_{\rm min} \approx 88.55^\circ, \qquad \chi \approx 2.76 \times 10^{-4}. \tag{9}$$

The full principal-angle spectrum at the biological point is (88.55°, 89.22°, 89.91°). This is

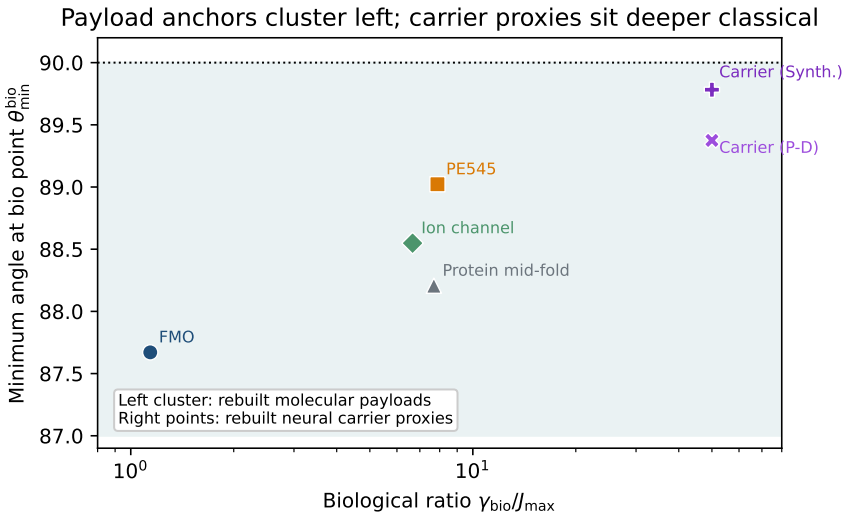

Figure 1: Biological operating points for the anchor set. The molecular payload anchors — FMO, PE545, the ion-channel point, and the protein mid-fold point — cluster in the same near-boundary band, with $\theta_{\min}^{\text{bio}}$ between roughly 87.7° and 89.0°. Two neural carrier proxies sit farther right at $\gamma/J_{\max} = 50$ and closer to the classical limit. The visual split is therefore the paper's central one: functional or candidate payload systems remain small-but-nonzero in χ, while the carrier layer is already substantially more classical on the same geometry map.

exactly the regime the paper needs: not strongly quantum, not fully classical, but slightly non-classical in a geometrically explicit sense.

The role of the model is intentionally narrow: it establishes a concrete boundary-near neural payload anchor. That is exactly the regime needed by Proposition 3.3. If each channel carries only a very small local load, the aggregate payload can still be large whenever many channels are entrained and redundancy is not overwhelming. For the neural application, this is the most direct route from local geometry to system-level dimensionality.

A compact robustness sweep makes the same point without overselling the model. Varying the nearest-neighbor coupling over $J_{\max} \in \{15, 20, 30, 40\}\,\text{cm}^{-1}$ and the dephasing over $\gamma \in \{100, 150, 200, 250, 300\}\,\text{cm}^{-1}$ keeps the biological-readout geometry in the same band: across the entire 20-point grid, $\theta_{\min}$ stays between 86.43° and 88.93°, while χ stays between 1.53×10^{-4} and 1.55×10^{-3}. Weaker coupling pushes the payload somewhat farther from the classical simplex; stronger coupling pulls it closer. But the sweep does not produce a collapse into either a strongly quantum or exactly classical regime. For the present thesis, that is the right robustness statement: the ion-channel payload anchor is parameter-sensitive, but not a one-point artifact.

The same is true at the level of minimal topology changes. Holding $J_{\max} = 30\,\text{cm}^{-1}$ and $\gamma_{\text{bio}} = 200\,\text{cm}^{-1}$ fixed, but varying the inner-well asymmetry over $\Delta \in \{10, 15, 20\}\,\text{cm}^{-1}$ and adding a next-nearest coupling $J_2 \in \{0, 5, 10\}\,\text{cm}^{-1}$, keeps the biological-point geometry tightly clustered: $\theta_{\min}$ remains between 88.55° and 88.77°, while χ remains between 2.20×10^{-4} and 2.77×10^{-4}. So even within the deliberately simple 4-site family, the neural payload anchor is not tied to one exact chain topology.

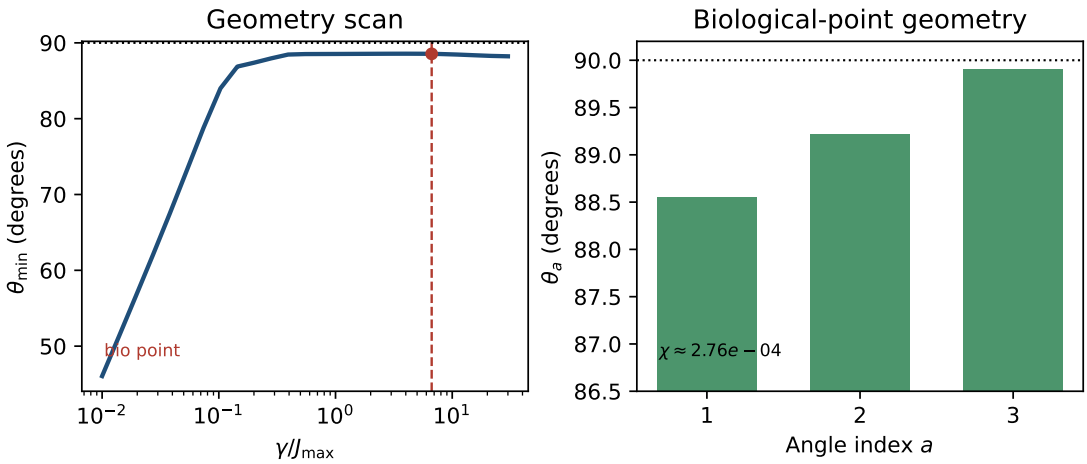

Figure 2: Minimal ion-channel payload anchor. Left: the minimum principal angle $\theta_{\min}$ for the KcsA-like 4-site model as a function of $\gamma/J_{\max}$, with the biological point marked. Right: the three Bures principal angles at the biological point. The geometric result is the point of the figure: at $\gamma_{\text{bio}}/J_{\max} = 6.67$, the ion-channel payload remains boundary-near with $\theta_{\min} \approx 88.55°$ and $\chi \approx 2.76 \times 10^{-4}$.

4.3 Protein microdomains as secondary payloads

Protein-folding microdomains, ligand-binding pockets, and other conformational substructures remain secondary in this paper. To represent that class of molecular payloads explicitly, we use an illustrative coarse-grained $d = 6$ mid-fold network,

$$H_{\text{fold}} = \begin{pmatrix} 20 & 25 & 12 & 0 & 0 & 0 \\ 25 & 35 & 18 & 22 & 0 & 0 \\ 12 & 18 & 42 & 20 & 10 & 0 \\ 0 & 22 & 20 & 50 & 24 & 12 \\ 0 & 0 & 10 & 24 & 38 & 26 \\ 0 & 0 & 0 & 12 & 26 & 28 \end{pmatrix}, \tag{10}$$

with initial site 0, target site 5, $\kappa = 0.5\,\text{ps}^{-1}$, and illustrative biological dephasing $\gamma_{\text{bio}} = 200\,\text{cm}^{-1}$. This is not a structure-specific folding Hamiltonian; it is a coarse molecular payload model intended to place a mid-fold microdomain in the same energy-scale analysis as the ion-channel anchor and the photosynthetic complexes. The resulting values are $J_{\max} = 26\,\text{cm}^{-1}$ and therefore $\gamma_{\text{bio}}/J_{\max} \approx 7.69$. At the biological point the same Lindblad-plus-geometry pipeline gives $\theta_{\min} \approx 88.21°$ and $\chi \approx 3.66 \times 10^{-4}$, with a local load $L_{\text{fold}} \approx 1.10 \times 10^{-2}$. So the folding microdomain lands in the same near-classical but non-classical region as the ion-channel payload. That broadens the payload story beyond excitability while preserving the present paper's hierarchy: ion channels remain the primary neural route, protein microdomains are the secondary generalization [24].

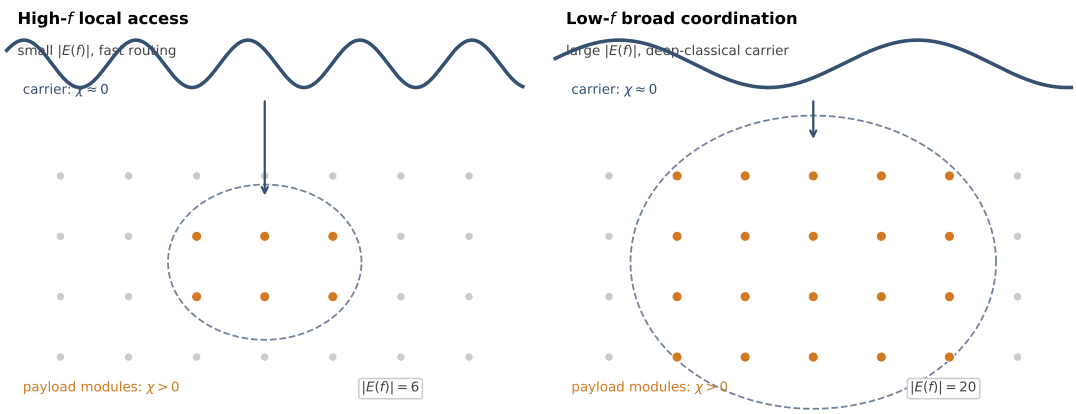

Figure 3: Conceptual carrier-payload inversion. Left: a faster carrier can still serve local access or routing while coordinating only a relatively small payload set. Right: a slower, more classical carrier can coordinate a much larger set of slightly non-classical molecular payload modules. The claim of the paper is about the size and geometry of the coordinated payload $E(f)$, not about the internal complexity of the carrier waveform itself.

5 Low-Frequency Coordination and Intelligence

5.1 Classical carriers, non-classical payloads

Neural oscillations at behavioral timescales are expected to be deep classical carriers. At those scales, dephasing dominates so strongly that the carrier itself should have $\chi_{\text{carrier}} \approx 0$. Two reduced carrier proxies confirm the direction of that limit. For a synthetic $d = 10$ gamma microcircuit, $\theta_{\min} \approx 89.43°$ with $\chi \approx 3.29 \times 10^{-5}$ already at $\gamma/J_{\max} = 1$, and $\theta_{\min} \approx 89.78°$ with $\chi \approx 3.72 \times 10^{-6}$ by the reference point $\gamma/J_{\max} = 50$. For a reduced Potjans–Diesmann cortical-column proxy ($d = 8$; based on the cell-type specific microcircuit of [25]), the same scan gives $\theta_{\min} \approx 88.27°$ with $\chi \approx 2.63 \times 10^{-4}$ at $\gamma/J_{\max} = 1$, and $\theta_{\min} \approx 89.37°$ with $\chi \approx 2.83 \times 10^{-5}$ by $\gamma/J_{\max} = 50$. So even the less classical of the two carriers still sits roughly $10\times$ below the ion-channel payload anchor in χ at the conservative reference point. This is not a defect in the argument; it is the feature that makes the carrier useful. A deep-classical carrier can propagate, synchronize, and entrain without adding significant geometric mixing of its own. That is why low-frequency waves are candidates for large-scale control rather than obstacles to complexity.

The disagreement with a frequency-centric high-frequency/high-dimensional reading can now be stated cleanly. High-frequency bursts may indeed track local access or flexible routing [1, 2]. This paper does not deny that. It denies the stronger inference that frequency itself is where dimensionality lives. Dimensionality resides in the coordinated substrate. If lower-frequency waves entrain larger cortical territories and therefore larger ion-channel payload sets [19], then Corollary 3.4 implies a higher coordinated dimensionality even when the carrier is more classical.

5.2 What "low frequency = high dimensional" means

The slogan needs to be read precisely. Figure 3 gives the qualitative picture; the equation below states it precisely. It does *not* mean that a low-frequency oscillation is internally more

complicated than a high-frequency oscillation. It means:

$$f \downarrow \implies |E(f)| \uparrow \implies D_{\text{coord}}(E(f)) \uparrow, \tag{11}$$

under the stated payload and redundancy assumptions.

This reframes the intelligence question. If cognition depends on the coordinated manipulation of many mixed-selectivity degrees of freedom [3, 4], then one should look for the geometry of the coordinated substrate, not just the spectral properties of the carrier. Low-frequency activity can be the right place to look for intelligence not because it is itself non-classical, but because it may be the most effective large-scale organizer of an ion-channel-rich payload that is not fully classical. Protein microdomains extend the same logic beyond excitability, but they are secondary to the ion-channel route for the present argument.

5.3 A minimal soft phase-coupling carrier model

The remaining question is how frequency enters the picture dynamically rather than rhetorically. A more biophysical bridge than the old hard-threshold story is a *soft phase-coupling* model. Let a carrier of frequency f propagate with speed v and amplitude envelope

$$A(r) = A_0 \exp\left(-\frac{r}{\lambda}\right), \tag{12}$$

where r is distance from the source and λ is an amplitude attenuation length. In the weighting model below we set $R_{\text{amp}} = \lambda$, so the amplitude scale appearing in the kernels is just the attenuation length of the carrier field. Instead of classifying a payload module as entrained or not entrained, we assign it a continuous coupling weight. Let

$$w_A(r) = \exp\left[-\left(\frac{r}{R_{\text{amp}}}\right)^{\eta}\right] \tag{13}$$

be an amplitude-decay kernel and

$$w_\phi(r, f) = \exp\left[-\left(\frac{r}{R_\phi(f)}\right)^{\eta}\right], \qquad R_\phi(f) = \frac{v\,\Delta\phi_{\max}}{2\pi f}, \tag{14}$$

be a phase-compatibility kernel. Here $\eta \geq 1$ controls how sharply coupling falls with distance, R_{amp} is the amplitude-decay length, and $R_\phi(f)$ is the familiar phase-limited coordination radius. Their product is

$$w(r, f) = w_A(r) w_\phi(r, f) = \exp\left[-\left(\frac{r}{R_{\text{eff}}^{(\eta)}(f)}\right)^{\eta}\right], \tag{15}$$

with soft-min coordination radius

$$R_{\text{eff}}^{(\eta)}(f) = \left(R_{\text{amp}}^{-\eta} + R_\phi(f)^{-\eta}\right)^{-1/\eta}. \tag{16}$$

If payload modules populate an effective spatial support of dimension s with spatial density ν_m, then the weighted payload size is

$$\widetilde{N}(f) = \nu_m C_s \Gamma\left(1 + \frac{s}{\eta}\right) \left(R_{\text{eff}}^{(\eta)}(f)\right)^{s}, \tag{17}$$

where $C_s = \pi^{s/2}/\Gamma(s/2+1)$ is the volume of the unit s-ball. In the phase-limited regime $R_\phi(f) \ll R_{\text{amp}}$, this still gives $\widetilde{N}(f) \propto f^{-s}$; at sufficiently low frequency the growth saturates smoothly at the amplitude ceiling instead of hitting a hard cutoff.

Proposition 5.1 (Soft phase-coupling scaling with saturation)**.** *Assume the soft phase-coupling model above, a common payload dimension d_m, and the nonzero-density/bounded-redundancy conditions of Proposition 3.3. Then*

$$D_{\text{coord}}(E(f)) \geq \nu_m C_s \Gamma\left(1+\frac{s}{\eta}\right)\left(R_{\text{eff}}^{(\eta)}(f)\right)^s \left[(d_m-1)+\frac{pL_\star}{\varrho_\star}\right] \tag{18}$$

In the phase-limited regime, coordinated dimensionality grows as f^{-s} as frequency falls; in the amplitude-limited regime, the growth saturates. In both cases, the low-frequency advantage comes from coordinating a larger payload set, not from the carrier itself becoming less classical.

Proof. Equation (17) gives the weighted payload size under the soft phase-coupling model. For modules of common dimension d_m, Definition 3.2 gives

$$D_{\text{coord}}(E(f)) = \widetilde{N}(f)(d_m-1) + \mathcal{L}_{\text{eff}}(E(f)).$$

Applying Proposition 3.3 to the weighted entrained set yields

$$\mathcal{L}_{\text{eff}}(E(f)) \geq \frac{pL_\star}{\varrho_\star}\widetilde{N}(f),$$

which gives the stated bound after substitution. □

Proposition 5.1 is still only a toy bridge. It does not fit real cortical attenuation, conduction anisotropy, or laminar structure. Its role is narrower and sufficient: it makes the paper's inversion explicit in a model with continuous coupling, finite propagation, and smooth saturation. Frequency enters through the admissible phase-lag radius. Dimensionality enters through the payload geometry being coordinated.

A reasonable objection is that any explicit radius-based construction could still be generating the low-frequency advantage by fiat. To check that, Figure 4 also overlays a heterogeneous soft-field simulation. We distribute 12,000 payload modules across a 2D disc of radius $6R_0$, assign lognormally distributed amplitude-decay lengths $R_{\text{amp},i}$ around $4R_0$ and baseline phase radii $R_{\phi,i}(f_0)$ around R_0 (log-width 0.35), and weight module i by

$$w_i(f) = \exp\left[-\left(\frac{r_i}{R_{\text{amp},i}}\right)^\eta - \left(\frac{r_i}{R_{\phi,i}(f)}\right)^\eta\right].$$

The simulation is normalized so that $\widetilde{N}(f_0) = N_0$. The resulting payload load tracks the analytic soft bridge near $f \sim f_0$, remains monotone as frequency falls, and still gives an approximately 1.4×10^2 gain in coordinated payload between $f/f_0 = 4$ and 0.25 at $\varrho = 1$ over the plotted range. So the inversion is not an artifact of a hard cutoff: it survives distributed tolerances, continuous weighting, and finite support.

A coarse sensitivity sweep makes that claim materially stronger. In the analytic bridge, varying $s \in \{1,2,3\}$, $\eta \in \{1,2,4\}$, and $R_{\text{amp}}/R_0 \in \{2,4,8\}$ keeps the low-frequency gain monotone in every case and produces gain factors between 6 and 3.9×10^3 across the same $f/f_0 = 4 \to 0.25$ range. For the heterogeneous 2D field check (so $s = 2$ is fixed by construction), varying η and R_{amp}/R_0 over the same ranges again keeps the gain monotone and yields factors

Band	f (Hz)	f/f_0	R_ϕ/R_0	$\widetilde{N}_{\text{hetero}}/N_0$	$\mathcal{L}_{\text{eff}}^{\text{hetero}}(\varrho = 10)$
Gamma	40	1.00	1.00	1.00	3.31
Beta	20	0.50	2.00	3.17	10.51
Theta	6	0.15	6.67	10.29	34.09
Delta	2	0.05	20.0	13.28	44.02
Infraslow	0.5	0.0125	80.0	13.78	45.68

Table 2: Band-level interpretation of the default heterogeneous soft phase-coupling bridge, using the ion-channel anchor with $N_0 = 10^4$, $R_{\text{amp}}/R_0 = 4$, $\eta = 2$, and redundancy factor $\varrho = 10$. The gain saturates smoothly below theta because the amplitude ceiling, not phase lag, becomes the limiting scale.

between roughly 35 and 325. The exact magnitude is model-dependent, as it should be, but the qualitative inversion is not confined to the default choice $(s, \eta, R_{\text{amp}}/R_0) = (2, 2, 4)$.

The same bridge is easier to read when expressed in familiar carrier bands. Taking $f_0 = 40\,\text{Hz}$ as a gamma-band reference, the default heterogeneous soft-field model gives about a $3.17\times$ coordinated-payload gain at beta (20 Hz), a $10.29\times$ gain at theta (6 Hz), and then smooth saturation through delta and infraslow frequencies as the amplitude ceiling takes over. So the present model does not predict an unlimited low-frequency divergence. It predicts a finite, saturating coordination advantage as the carrier slows.

Using the ion-channel anchor makes the scale of the effect explicit. For the 4-site selectivity-filter model,

$$L_{\text{ion}} = \chi(d^2 - d) \approx (2.76 \times 10^{-4}) \times 12 \approx 3.31 \times 10^{-3} \tag{19}$$

per channel at the biological point. If one takes an illustrative baseline of $N_0 = 10^4$ weighted channels at $f = f_0$ in the phase-limited regime, then the soft bridge gives

$$\mathcal{L}_{\text{eff}}(f) \approx \frac{N_0 L_{\text{ion}}}{\varrho} \left(\frac{R_{\text{eff}}^{(\eta)}(f)}{R_{\text{eff}}^{(\eta)}(f_0)} \right)^s, \qquad R_0 := R_\phi(f_0). \tag{20}$$

Figure 4 plots this for $s = 2$, $R_{\text{amp}}/R_0 = 4$, $\eta = 2$, and redundancy factors $\varrho \in \{1, 10, 100\}$. The point is not the exact numerical choice of N_0 or R_{amp}/R_0; it is that a very small local non-classical load can become system-relevant once multiplied by large coordinated site counts, while the low-frequency gain saturates rather than diverging forever.

There is also a clean way to state the fractional size of the effect. For homogeneous payload modules,

$$\Phi := \frac{\mathcal{L}_{\text{eff}}}{N(d_m - 1)} = \frac{\chi(d_m^2 - d_m)}{\varrho(d_m - 1)} = \frac{\chi d_m}{\varrho}. \tag{21}$$

Thus the aggregate non-classical payload is extensive in N, but the fractional uplift above the classical baseline is independent of N for fixed (d_m, χ, ϱ). For the ion-channel anchor ($d_m = 4$, $\chi \approx 2.76 \times 10^{-4}$),

$$\Phi_{\text{ion}} \approx \frac{1.10 \times 10^{-3}}{\varrho}, \tag{22}$$

so the toy bridge says two things simultaneously: the per-channel effect is small, and the absolute payload load still scales linearly with the number of entrained channels. Whether a fractional uplift of this size is functionally amplified by real neural dynamics is an empirical question, not a claim already established here. The present point is narrower: the uplift is nonzero, geometrically structured, and extensive in coordinated site count.

N	Classical $3N$	$\mathcal{L}_{\text{eff}}$ $(\varrho = 1)$	$\mathcal{L}_{\text{eff}}$ $(\varrho = 10)$	$\mathcal{L}_{\text{eff}}$ $(\varrho = 100)$
10^3	3.0×10^3	3.31	3.31×10^{-1}	3.31×10^{-2}
10^4	3.0×10^4	3.31×10^1	3.31	3.31×10^{-1}
10^5	3.0×10^5	3.31×10^2	3.31×10^1	3.31

Table 3: Ion-channel payload benchmarks at the biological point, using the anchor $L_{\text{ion}} \approx 3.31 \times 10^{-3}$ per channel. The corresponding fractional uplifts above the classical baseline are $\Phi_{\text{ion}} \approx 1.10 \times 10^{-3}$ for $\varrho = 1$, 1.10×10^{-4} for $\varrho = 10$, and 1.10×10^{-5} for $\varrho = 100$. The message is not that the fraction is large; it is that the absolute non-classical payload remains extensive in N.

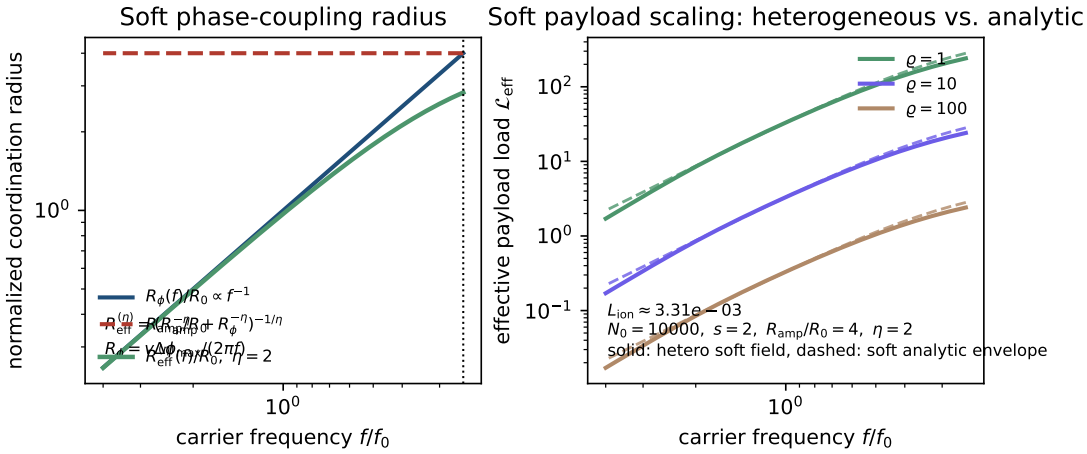

Figure 4: Soft phase-coupling bridge used in Section 5. Left: the phase-limited radius $R_\phi(f) = v\Delta\phi_{\max}/(2\pi f)$ and amplitude scale R_{amp} combine into a soft-min radius $R_{\text{eff}}^{(\eta)}(f) = (R_{\text{amp}}^{-\eta} + R_\phi(f)^{-\eta})^{-1/\eta}$. Lower-frequency carriers enlarge the phase-limited radius until smooth saturation at the amplitude scale. Right: using the actual ion-channel anchor from Figure 2, dashed curves show the analytic soft payload load $\mathcal{L}_{\text{eff}}(f) \approx N_0 L_{\text{ion}}(R_{\text{eff}}^{(\eta)}(f)/R_{\text{eff}}^{(\eta)}(f_0))^s/\varrho$ for illustrative $N_0 = 10^4$, $s = 2$, $R_{\text{amp}}/R_0 = 4$, $\eta = 2$, and redundancy factors $\varrho \in \{1, 10, 100\}$. Solid curves show a heterogeneous soft-field simulation with distributed amplitude scales and phase tolerances, normalized to match $\widetilde{N}(f_0) = N_0$. The low-frequency advantage and saturation structure survive continuous weighting and heterogeneity, so the effect is not created by a single hard threshold.

5.4 Predictions

The carrier-payload view yields direct predictions:

1. Measures of cognitive dimensionality should track entrained ion-channel payload diversity better than carrier frequency alone.

2. Manipulations that preserve large-scale oscillatory coherence but reduce the diversity or number of available molecular payload sites should reduce effective dimensionality.

3. If low-frequency rhythms have larger phase-tolerant coordination radii, then the coordinated dimensionality should scale with entrained territory size as in Proposition 5.1, not with frequency alone.

4. The right computational bottleneck is not whether classical surrogates exist in principle, but how expensive it is to preserve the effective payload $\mathcal{L}_{\text{eff}}$ rather than projecting it

Observable or manipulation	Frequency-dominant expectation	Carrier-payload expectation
Cross-condition variation in measured dimensionality	Dimensionality should increase primarily with carrier frequency or burstiness.	Dimensionality should increase primarily with entrained payload size and payload diversity, even when the carrier is slower.
Hold large-scale carrier coherence fixed while reducing available payload sites or payload diversity	Only modest changes in dimensionality are expected if the carrier spectrum is preserved.	Effective dimensionality should fall because the payload, not the carrier frequency, is where state-space richness resides.
Compare low-frequency broad-coordination states with high-frequency local-access states	High-frequency states should appear more dimensional because they are spectrally faster.	Low-frequency states can appear more dimensional when they coordinate a larger payload ensemble, even if high-frequency states remain better markers of local access or routing.

Table 4: Alternative empirical expectations. The paper does not deny that high-frequency bursts can correlate with flexible access or gating. It denies the stronger inference that carrier frequency itself is the primary locus of dimensionality.

away, even when the fractional uplift Φ is modest.

These predictions are useful only if they are contrasted with a simpler frequency-dominant interpretation. Table 4 states that contrast in the sharpest operational form available in the present draft.

The paper is testable without direct molecular-state tomography. In practice, χ_i and L_i can be treated as latent payload variables, while the model's observable consequences appear in conditional relationships between carrier coherence, coordinated territory size, payload diversity, and recorded neural dimensionality. The right empirical question is therefore not "is the carrier fast?" but "after controlling for carrier coherence, does measured dimensionality track the size and diversity of the coordinated payload better than it tracks carrier frequency alone?" Table 5 states that route in operational terms.

A minimal empirical test can be written directly at the level of recorded neural data:

$$D_{\text{obs}} = \beta_0 + \beta_1 \log \widetilde{N} + \beta_2 H_{\text{payload}} + \beta_3 C_{\text{carrier}} + \beta_4 \log f + \varepsilon, \tag{23}$$

where D_{obs} is an observed neural dimensionality measure, $\widetilde{N}$ is the coordinated territory or weighted payload size, H_{payload} is a payload-diversity proxy, and C_{carrier} is a carrier-coherence proxy. The carrier-payload picture predicts $\beta_1, \beta_2 > 0$ and, crucially, that β_4 should shrink toward zero once $\widetilde{N}$, H_{payload}, and C_{carrier} are included. The frequency-dominant interpretation predicts the opposite: frequency should remain a dominant positive predictor even after those controls are applied. That regression-level disagreement is the shortest empirical route to deciding between the two pictures.

6 Discussion

The main gain of the present reformulation is conceptual clarity. The old binary question—is biology quantum or classical?—is the wrong one. The geometric question is better: how much

Theoretical object	Practical proxy	Carrier-payload expectation
Carrier coherence / stability	Low-frequency phase locking, coherence radius, or cross-area synchrony	A more reliable carrier should improve dimensional coordination mainly by enlarging the territory over which a payload can be entrained, not by increasing dimensionality on frequency alone.
Entrained payload size $\lvert E(f)\rvert$	Spatial footprint of phase-locked tissue, laminar recruitment span, or effective coordination territory	Conditional on comparable payload statistics, larger coordinated territory should predict larger measured dimensionality.
Payload diversity / availability	Channel or receptor diversity proxies, pharmacological availability of conductance classes, or cell-type conductance richness	Reducing payload diversity at fixed carrier coherence should reduce dimensionality even if the carrier spectrum is preserved.
Observed coordinated dimensionality	Participation ratio, cross-validated manifold dimension, shattering dimension, or mixed-selectivity capacity [3, 4]	Dimensionality should be better explained by coordinated territory and payload diversity than by carrier frequency alone.

Table 5: Operational measurement route for the carrier-payload program. The proposal does not require direct measurement of molecular χ_i in the first instance. It requires testing whether recorded dimensionality is better predicted by coordinated territory and payload diversity than by carrier frequency alone.

of the coherence sector remains accessible, where is it located, and how is it coordinated across scales?

Three claims survive scrutiny in the current draft. First, decoherence is a dimensional projection of state space. Second, near-boundary biological dynamics are real, with photosynthesis as the proof of principle. Third, low-frequency carrier advantage is mathematically coherent once dimensionality is relocated to the coordinated payload and coupled to frequency through a transparent soft phase-coupling model.

Several things remain open. The current analysis includes reduced photosynthetic proof-of-principle anchors, two neural-relevant payload anchors, and two carrier proxies, but it does not yet span the broader family of molecular payload models or a data-constrained cortical carrier model. The redundancy factor ϱ_E needs empirical grounding rather than remaining a formal placeholder. And the neuroscience side still needs a genuinely data-constrained carrier-to-payload coupling model beyond the reduced soft phase-coupling bridge introduced in Section 5. Table 5 is therefore important: it marks the shortest path from the present theory to an empirical attack. Those are real gaps. They are also the right gaps: they are downstream of a coherent theory rather than upstream of one.

The paper therefore makes a narrower but more defensible claim than strong quantum-cognition proposals. It does not say that neural oscillations are quantum. It does not say that high dimensionality forces non-classicality. It says that biological systems built from many structured molecular machines can carry an extensive non-classical payload if a nonzero density of those machines sits near the quantum-classical boundary. Large-scale low-frequency waves may then be the right place to look for intelligence because they are the best classical carriers of that payload. In neural tissue, ion channels are the first place to test that claim. The numerical lesson of the draft is correspondingly sober: the local fractional uplift is small [Eq. (22)], but the absolute payload remains extensive in entrained site count, and whether that uplift is

functionally consequential is an empirical question for the measurement program in Table 5.

That is the full intended inversion of the usual intuition. Frequency is not dimensionality. Frequency is coordination. Dimensionality lives in the substrate being coordinated.

Data and Code Availability

All code used to generate the results and figures in this manuscript is available in the accompanying GitHub repository: github.com/todd866/decoherence-dimensional-collapse. The current draft is generated directly from the analysis scripts and result files in that repository.

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